Game Theory in Electricity Markets

Exercise Sheet 4 — Smoothness and the Price of Anarchy

2nd of October 2026

Exercise 1 Smoothness Bounds for Coarse Correlated Equilibria

In the lecture we proved the smoothness bound for pure Nash equilibria: if a cost-minimization game with social cost C is (λ, μ)-generalized smooth, i.e. for all $a, a' \in A$

$$\sum_{i=1}^{n} c_i(a'_i, a_{-i}) - \sum_{i=1}^{n} c_i(a) + C(a) \;\leq\; \lambda\, C(a') + \mu\, C(a),$$

with $\lambda > 0$ and $\mu < 1$, then every pure Nash equilibrium a satisfies $C(a) \leq \frac{\lambda}{1-\mu} \min_{a'} C(a')$. We also stated that the bound remains true for the other equilibrium concepts we have seen.

Task: Prove that the smoothness bound also holds for coarse correlated equilibria, i.e. show that every CCE $\pi \in \Delta(A)$ of a (λ, μ)-generalized smooth game satisfies

$$\mathbb{E}_{a\sim\pi}\big[C(a)\big] \;\leq\; \frac{\lambda}{1-\mu} \min_{a' \in A} C(a').$$

The proof follows the same arguments as the one for pure Nash equilibria. It is an especially good exercise for everybody who wants to practice handling distributions and expected values.

Exercise 2 Smoothness in the Prisoner's Dilemma

Remember the prisoner's dilemma. Two players $1, 2$ simultaneously choose between C and D, and each of them wants to *minimize* their own cost. The costs (c_1, c_2) are given by

		Player 2	
		C	D
Player 1	C	(2, 2)	(10, 0)
	D	(0, 10)	(9, 9)

with social cost $C(a) = c_1(a) + c_2(a)$.

(2.1) Find the pure Nash equilibrium and the socially optimal action profile, and compute the price of anarchy of the Nash equilibrium.

(2.2) Calculate the best smoothness bound $\inf\{\frac{\lambda}{1-\mu}\}$ for this game, e.g. by solving the linear program from the lecture.

(2.3) Make a smoothness sandwich: compare the smoothness bound with the price of anarchy of the Nash equilibrium. What do you observe?

Exercise 3 Smoothness Bounds via Linear Programming

Recall the curtailment game from Exercise 3 of last week's sheet: four wind farms $W_1, \ldots, W_4$ choose $a_i \in \{F, C\}$ (feed in or curtail), and $k(a)$ denotes the number of wind farms that choose F. Since

smoothness is formulated for cost-minimization games, we turn the payoffs u_i of last week into nonnegative costs

$$c_i(a) = 10 - u_i(a), \qquad C(a) = \sum_{i=1}^{4} c_i(a).$$

(3.1) Formulate the linear program that gives the best smoothness bound $\inf\{\frac{\lambda}{1-\mu}\}$ over all feasible (λ, μ). Note that $\frac{\lambda}{1-\mu}$ is not linear in (λ, μ), so you might want to use the change of variables $\rho = \frac{\lambda}{1-\mu}$, $\nu = \frac{1}{1-\mu}$ from the lecture.

(3.2) Formulate the linear program that gives the smoothness bound when we only compare against the socially optimal action profile $a^* = (F, F, C, C)$, i.e. $a' = a^*$ is fixed in the smoothness condition.

(3.3) Solve both linear programs numerically, with a solver of your choice, and report the two bounds.

(3.4) Calculate PoA_{CCE} for this game (it is not the same as last week as the games are slightly different). Compare the three numbers with each other. Which relations between the numbers must hold, and which ones are specific to this game?

Exercise 4 Smoothness in an EV Charging Game

Consider n electric vehicles (or, more generally, price-responsive flexible loads) that each have to charge in exactly one of two time slots $t \in \{1, 2\}$, for example night and day. Let $x_t(a)$ denote the number of vehicles that charge in slot t under the action profile a. The price in a slot is proportional to its load, so every vehicle that charges in slot t pays

$$c_i(a) = x_t(a) \quad \text{(EUR)},$$

and the social cost is $C(a) = \sum_i c_i(a) = x_1(a)^2 + x_2(a)^2$.

(4.1) For what follows recall/observe that for all integers $x, y \geq 0$ it holds that

$$y\,(x+1) \;\leq\; \tfrac{5}{3}\,y^2 + \tfrac{1}{3}\,x^2.$$

(4.2) Use (4.1) to show that the game is $(\frac{5}{3}, \frac{1}{3})$-generalized smooth, for any number of vehicles n. Conclude that $\text{PoA} \leq \frac{5}{2}$ for pure Nash equilibria and, using Exercise 1, also for coarse correlated equilibria.

(4.3) Set $n = 4$. Compute the optimal social cost and the worst-case social cost over all CCE with the LP approach from last week, and compute the best smoothness bound with the LP from Exercise 2. How tight is the bound $\frac{5}{2}$ from (4.2) for this instance?

Solution of (4.2). Let $a, a' \in A$ be arbitrary profiles, and let $x_t = x_t(a)$ and $y_t = x_t(a')$ denote the number of vehicles charging in slot $t \in \{1, 2\}$ under a and a', respectively.

Step 1 (deviation cost). Consider vehicle i deviating from a_i to $a'_i = t$ while all others keep playing a_{-i}. If $a_i \neq t$, the load in slot t increases to $x_t + 1$, so $c_i(a'_i, a_{-i}) = x_t + 1$. If $a_i = t$, nothing changes and $c_i(a'_i, a_{-i}) = x_t \leq x_t + 1$. In both cases

$$c_i(a'_i, a_{-i}) \;\leq\; x_{a'_i} + 1.$$

Step 2 (summation). Under a' exactly y_t vehicles choose slot t. Grouping the vehicles by their slot in a' gives

$$\sum_{i=1}^{n} c_i(a'_i, a_{-i}) \;\leq\; \sum_{i=1}^{n} \left(x_{a'_i} + 1\right) \;=\; \sum_{t=1}^{2} y_t\,(x_t + 1).$$

Step 3 (apply (4.1)). Since x_t, y_t are nonnegative integers, (4.1) yields

$$\sum_{t=1}^{2} y_t\,(x_t + 1) \;\leq\; \frac{5}{3}\sum_{t=1}^{2} y_t^2 + \frac{1}{3}\sum_{t=1}^{2} x_t^2 \;=\; \frac{5}{3}\,C(a') + \frac{1}{3}\,C(a).$$

Step 4 (smoothness). Because $C(a) = \sum_i c_i(a)$, the left-hand side of the generalized smoothness condition simplifies to

$$\sum_{i=1}^{n} c_i(a_i', a_{-i}) - \sum_{i=1}^{n} c_i(a) + C(a) \;=\; \sum_{i=1}^{n} c_i(a_i', a_{-i}).$$

Together with Steps 1–3 we obtain, for all $a, a' \in A$,

$$\sum_{i=1}^{n} c_i(a_i', a_{-i}) - \sum_{i=1}^{n} c_i(a) + C(a) \;\le\; \frac{5}{3}\, C(a') + \frac{1}{3}\, C(a),$$

i.e. the game is (λ, μ)-generalized smooth with $\lambda = \frac{5}{3} > 0$ and $\mu = \frac{1}{3} < 1$, independently of n.

Step 5 (price of anarchy). By the smoothness theorem, every pure Nash equilibrium satisfies

$$\mathrm{PoA} \;\le\; \frac{\lambda}{1-\mu} \;=\; \frac{5/3}{2/3} \;=\; \frac{5}{2}.$$

By Exercise 1 the same bound holds for every coarse correlated equilibrium, and hence also for all equilibrium concepts contained in it. □